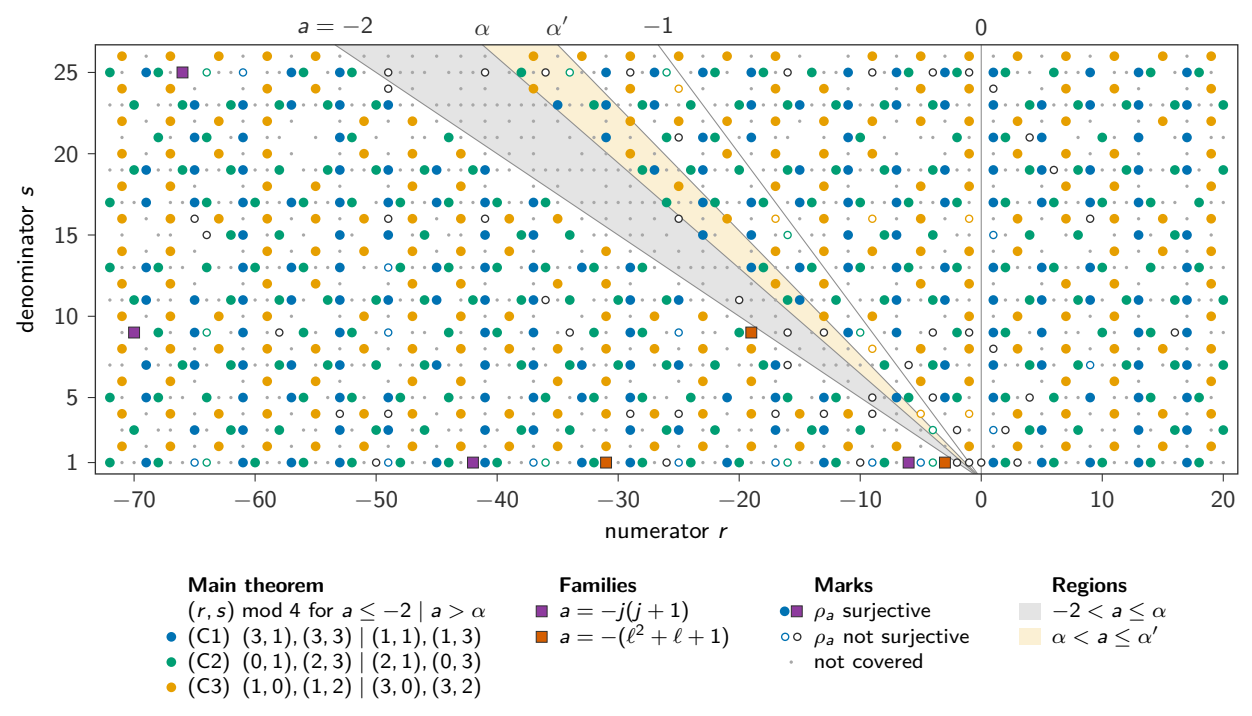

$a = -2$
α
α'
-1
0
denominator s
numerator r
25
20
15
10
5
1
−70
−60
−50
−40
−30
−20
−10
0
10
20
Main theorem
(r, s) mod 4 for $a \le -2 \mid a > \alpha$
(C1) (3, 1), (3, 3) | (1, 1), (1, 3)
(C2) (0, 1), (2, 3) | (2, 1), (0, 3)
(C3) (1, 0), (1, 2) | (3, 0), (3, 2)
Families
$a = -j(j+1)$
$a = -(\ell^2 + \ell + 1)$
Marks
ρ_a surjective
ρ_a not surjective
not covered
Regions
$-2 < a \le \alpha$
$\alpha < a \le \alpha'$